

Part 1 - Pipeline Execution Diagram (Lines 3-16)

The diagram below traces the 5-stage pipeline execution (IF, ID, EX, MEM, WB) for lines 3 through 16 of the Bubble Sort algorithm.

- **Stall Cycles (**):** Pipeline bubbles inserted due to RAW flag hazards or load-use delays where IF/ID is frozen.
- **Flushed Cycles (XX):** Instructions in IF/ID discarded due to taken branches or jumps.

Line	Instruction	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
L3	CMP A, D	IF	ID	EX	MEM	WB															
L4	BRGE End		IF	ID	**	EX	MEM	WB													
L5	LOAD D, [last]			IF	**	ID	EX	MEM	WB												
L6	SUB D, A				**	IF	ID	EX	MEM	WB											
L7	CMP B, D						IF	ID	EX	MEM	WB										
L8	BRGE Inc							IF	ID	**	EX	MEM	WB								
L9	LOADF C, [arr+B]								IF	**	ID	EX	MEM	WB							
L10	LOADF D, [arr+B+1]									**	IF	ID	EX	MEM	WB						
L11	CMP D, C											IF	ID	EX	MEM	WB					
L12	BRGE Jinc												IF	ID	**	EX	MEM	WB			
L13	STOREF [arr+B], D													IF	**	XX	-	-	-	-	-
L15	ADDI B, 1														**	IF	ID	EX	MEM	WB	
L16	JUMP Inner																IF	ID	EX	MEM	WB
L7	CMP B, D (Target)																	IF	XX	-	-

Part 2 — Hazard Analysis Table

The table below summarizes every data, flag, and control hazard present in the trace above, along with its cause and resolution scheme.

Instruction Pair	Hazard Type	What Causes It	Resolution Strategy
L3 → L4	Flags RAW	CMP writes flags in EX stage. BRGE in ID evaluates flags before they reach the register file.	1-Cycle Stall + Flags Forwarding: Insert 1 bubble (**), freeze IF/ID, then forward flags from EX/MEM to ID.
L7 → L8	Flags RAW	CMP updates arithmetic flags in EX; conditional branch reads flags in ID.	1-Cycle Stall + Flags Forwarding: Insert 1 bubble (**) and forward EX/MEM.flags.
L9 → L11	Load-Use RAW	LOADF writes register C in WB, but CMP reads C in EX 2 cycles later.	Forwarding (MEM/WB → EX): Forward memory output from MEM/WB directly to ALU input B in EX. No stall required.
L11 → L12	Flags RAW	CMP updates flags in EX stage; BRGE evaluates condition in ID.	1-Cycle Stall + Flags Forwarding: Stall 1 cycle (**) and forward EX/MEM.flags to branch logic.
L12 → L13	Control	BRGE evaluates as Taken in EX. L13 (STOREF) was speculatively fetched in IF.	Flush IF/ID: Overwrite IF/ID with a NOP bubble (XX) to discard L13 and redirect PC to L15 (Jinc).
L16	Control	Unconditional jump resolves target address in EX stage.	Flush IF/ID: Always flush the IF/ID register (XX) on JUMP and update PC with target address.